INT8 Post-Training Quantization and Object Size in Aerial Object Detection

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***Abstract*—INT8 post-training quantization is a common way to make object detectors fast enough for on-board use in drones, and its accuracy cost is usually reported as one aggregate value. Aerial images, however, are dominated by small objects, which are often assumed to be more sensitive to reduced numerical precision. This study tests that assumption with a pre-registered experiment. A YOLOv8n detector was fine-tuned on VisDrone2019-DET and converted with TensorRT into FP32, FP16 and INT8 engines, using three different INT8 calibration sets. All engines were evaluated on the same 1,610 test-dev images, and the difference in relative average precision loss between small and non-small objects was estimated with a paired image-level bootstrap. INT8 reduced average precision by 6.43 percent for small objects and by 5.47 percent for non-small objects. The difference of 0.96 percentage points had a 95 percent confidence interval from minus 0.81 to plus 3.03 points. Because the interval includes zero, the pre-registered result is inconclusive, and the data do not show that INT8 harms small objects more than larger ones. The difference was positive for all three calibration sets, but none of the intervals excluded zero. The data rule out a size-dependent penalty larger than about three percentage points, which is moderate compared with the overall INT8 loss of 5.4 percent.**

***Index Terms*—aerial imagery, bootstrap, INT8, object detection, post-training quantization, small objects, TensorRT**

I. Introduction

Unmanned aerial vehicles (UAVs) increasingly run object detectors on board, where memory, energy and latency budgets are tight. A standard way to meet such budgets is post-training quantization (PTQ) to 8-bit integers (INT8). A network trained in 32-bit floating point (FP32) is converted after training, using a small set of calibration images to estimate the value ranges of its activations; no retraining is needed [1], [2]. Deployment toolchains such as TensorRT make this conversion a routine export step for YOLO detectors [3]. Its accuracy cost is typically reported as a single change in mean average precision (mAP).

A single overall accuracy score can hide how quantization affects different groups of objects differently. In image classification, compressed networks lose accuracy disproportionately on a subset of examples rather than uniformly across them [4]. In aerial detection, the subset of concern is small objects. They make up most annotations in drone benchmarks such as VisDrone [5], [6] and are already the hardest to detect [7]. Two mechanisms make a larger INT8 penalty for small objects plausible. First, a small object is represented by few activations, and their resolution can suffer when the quantization range is dominated by large outlier values, a known difficulty of PTQ for detectors [2], [8]. Second, a localization error of a single pixel lowers the intersection over union (IoU) of a small box much more than that of a large box.

If small objects do lose more, aggregate mAP overstates how suitable INT8 is for aerial deployment, where small objects often matter most. If they do not, size-specific countermeasures are unnecessary. Detection benchmarks commonly report accuracy by COCO size group [9]. However, the PTQ studies cited here focus on proposing methods rather than on testing whether the loss depends on object size. The question addressed here is therefore the following.

Research question: *In aerial imagery, is the relative loss in detection accuracy caused by INT8 post-training quantization of a lightweight YOLO detector larger for small objects than for larger objects?*

The independent variable is numerical precision: an FP32 baseline versus an INT8 treatment, with an FP16 control. Object size (small vs. non-small) is an observed moderator, and the dependent variable is AP@[.5:.95] per size group. With D defined in Section III as the relative AP loss of small objects minus that of non-small objects, the hypotheses are $H_0 : D \leq 0$ and $H_1 : D > 0$.

The study is deliberately narrow: one detector, one dataset and one quantization toolchain, so that numerical precision is the only manipulated factor. Its contributions are (i) a pre-registered, paired experimental design in which all precision conditions derive from one frozen checkpoint and are evaluated on identical images, (ii) a paired image-level bootstrap for COCO-style AP by object size, and (iii) an estimate, with its uncertainty, of the size-dependence of INT8 PTQ loss on VisDrone2019-DET. Section II reviews related work, Section III describes the method, Section IV reports the results, and Sections V and VI discuss and conclude.

II. Related Work

Quantization. Integer-arithmetic inference with affine quantization of weights and activations was formalized by Jacob et al. [1]. Practical PTQ, including per-channel weight quantization and calibration of activation ranges, is reviewed by Nagel

et al. [2] and Krishnamoorthi [10]. Wu et al. [11] compare INT8 calibration methods, including the max-based calibration used here, and show that INT8 accuracy can be kept within 1% of FP32 for all networks studied; as in most PTQ work, accuracy is reported as one aggregate metric per model. Low precision can also be used during training, as in unified INT8 training by Zhu et al. [12]. Such approaches change the trained model and are outside the scope of this study.

Quantization of detectors. Kim et al. [8] study PTQ specifically for object detection models and address the influence of outlier activations on calibration. Like most work in this area, they propose a quantization method. The present study instead measures the behaviour of a standard deployment toolchain [3].

Small objects in aerial imagery. VisDrone [5] is a widely used benchmark for detection in drone imagery, and the VisDrone-DET2021 challenge results [6] document how difficult small, densely packed objects remain. Akyon et al. [7] address small objects through slicing-aided inference, which highlights the role of object scale. The COCO small/medium/large definition [9] is the standard way to report accuracy by size.

Non-uniform effects of compression. Hooker et al. [4] showed that pruned and quantized classifiers lose accuracy disproportionately on a subset of examples. Whether an analogous, size-related subset exists for detection under INT8 PTQ in aerial imagery is the question investigated here.

III. Method

A. Study Design

This study examines whether INT8 post-training quantization causes a larger relative detection-accuracy loss for small objects than for larger objects in aerial imagery. Precision is the experimental factor, with a TensorRT FP32 engine as the baseline, an INT8 engine as the primary treatment, and an FP16 engine as a control. The FP16 control is needed because the toolchain runs the non-quantized layers of the INT8 engine in FP16. Object size is an observed moderator. The comparison is paired: all precision conditions are evaluated on the same images. The design, metrics, size groups and analysis were fixed in a pre-registration dated 7 Oct. 2026, before any model was trained. It is published with the code and a deviation log [13].

For each object-size group g, relative AP loss is defined as

$$\mathrm{RelDrop}_g = \frac{\mathrm{AP}_g^{\mathrm{FP32}} - \mathrm{AP}_g^{\mathrm{INT8}}}{\mathrm{AP}_g^{\mathrm{FP32}}}. \tag{1}$$

The primary contrast is

$$D = \mathrm{RelDrop}_{\mathrm{small}} - \mathrm{RelDrop}_{\mathrm{non\text{-}small}}. \tag{2}$$

The alternative hypothesis is $D > 0$, a greater relative quantization loss for small objects. The null hypothesis is $D \leq 0$. The relative loss is used because small objects start from a much lower AP. The same proportional harm therefore produces a smaller absolute drop, which would bias an absolute comparison against the alternative hypothesis.

B. Dataset and Annotation Processing

The experiment uses VisDrone2019-DET [5]. The training split is used for fine-tuning and INT8 calibration, the validation split for checkpoint selection and development, and test-dev only for the final evaluation. The splits contain 6,471 training, 548 validation and 1,610 test-dev images. Dataset archives are verified by SHA-256 before processing.

Annotations are converted to YOLO format for training and to COCO JSON for evaluation. VisDrone annotations with score zero, which comprise the ignored regions and the "others" category, are excluded. Bounding boxes are clipped to the image boundaries, and boxes with a width or height below one pixel after clipping are discarded. Detections on excluded ignored regions are therefore treated as false positives for every condition. As a result, absolute AP values are not directly comparable with the official VisDrone evaluation toolkit [6], although the paired comparison is unaffected.

Object size is the area of the clipped ground-truth box in original image pixels. Following the COCO convention [9], small objects have an area below $32^2 = 1{,}024$ pixels and non-small objects an area of at least $1{,}024$ pixels. The secondary groups are medium ($32^2 \leq \mathrm{area} < 96^2$) and large ($\mathrm{area} \geq 96^2 = 9{,}216$ pixels). After processing, test-dev contains 75,102 objects, of which 67.7% are small, 29.1% medium and 3.2% large (60.5%, 34.0% and 5.5% in the training split). Size and class are related: pedestrians, people and motors make up 58.5% of small but only 14.4% of non-small objects, whereas cars, vans, trucks and buses make up 38.3% and 82.1%, respectively. Size-specific evaluation follows the COCO rule: ground truths outside a size range, detections matched to them, and unmatched detections outside the range are ignored for that range. Unlike slicing-aided approaches [7], every precision condition uses the same full-image input; tiling is not part of the study.

C. Model Training and Engine Construction

The detector is YOLOv8n [14], the smallest YOLOv8 variant and a typical choice for embedded hardware, initialized from COCO-pretrained weights. One model is fine-tuned on the training split at 640×640 for 100 epochs, with batch size 16, random seed 0 and deterministic training. Other options use the Ultralytics defaults, and no hyper-parameter search is performed. The best checkpoint on the validation split is frozen and used to build all precision variants.

The frozen checkpoint is exported to ONNX with a static input of $1 \times 3 \times 640 \times 640$ and converted to TensorRT engines with the documented Ultralytics export workflow [3]. The exported ONNX graph is identical for all conditions (verified by hash) before the precision-specific conversion. Engines are built in FP32, FP16 and INT8. Three independent INT8 calibration sets are sampled, one per seed (0, 1 and 2), each consisting of 512 training images drawn uniformly at random. Calibration and evaluation use the same 640×640 letterbox preprocessing. The primary INT8 comparison uses calibration seed 0, and seeds 1 and 2 serve as robustness checks. For INT8, NVIDIA ModelOpt inserts explicit quantize/dequantize

nodes with max calibration, and layers that are not quantized run in FP16.

D. Evaluation Measures

The frozen PyTorch checkpoint and every TensorRT engine are evaluated with identical inference settings: confidence threshold 0.001, non-maximum suppression (NMS) IoU threshold 0.7, at most 500 detections per image, batch size 1, and square 640×640 input. The primary accuracy measure is COCO-style AP averaged over IoU thresholds from 0.50 to 0.95 (AP@[.5:.95]); AP at IoU 0.50 (AP@0.5) is secondary. Evaluation is reported overall and by size group.

Predictions are matched to ground truth with pycocotools, and the per-image match records are stored for the bootstrap. The implementation's AP recomputation must agree with pycocotools to within 10^{-9}. As further implementation checks, the PyTorch FP32 checkpoint is compared with the TensorRT FP32 engine, and the results are compared with Ultralytics' validation metrics. All development and debugging used the validation split. Each engine was evaluated on test-dev exactly once (12 Oct. 2026), after all engines had been built and evaluated on the validation split. The confirmatory comparison is TensorRT FP32 versus INT8 calibration seed 0 on test-dev.

E. Statistical Analysis

AP is computed from detections ranked across the whole test set, so there are no independent per-image AP values for a classical paired test. A paired nonparametric bootstrap [15] is therefore used, with the image as the resampling unit. It draws 2,000 resamples of the 1,610 test-dev images with replacement (random seed 0). The same resampled image indices are used for every precision condition and contrast, which preserves the pairing. For each resample, AP is recomputed by size group from the stored match records, without re-matching detections.

The confirmatory statistic is D in (2) for FP32 versus INT8 seed 0 on AP@[.5:.95], with a two-sided 95% percentile bootstrap confidence interval. If the lower bound is above zero, the null hypothesis is rejected in favour of a greater relative loss for small objects. If the interval contains zero, the result is inconclusive, and if the upper bound is below zero, the result is contrary to the alternative hypothesis. Because H_1 is directional, this rule corresponds to a one-sided test at $\alpha = 0.025$.

The percentile bootstrap makes no normality assumption about D, which is a ratio-based quantity with an unknown sampling distribution. It does assume that test images are independent and that the resampled AP is well defined. Both are checked: AP must be finite for every resample, the shape of the bootstrap distribution is inspected for skew and bias, and possible dependence between images of the same scene is discussed as a validity threat. The image is used as the unit because VisDrone test-dev file names do not reliably identify video sequences, so no sound cluster structure was available. This was decided from the labels alone, before any model output was produced.

TABLE I
TEST-DEV AP@[.5:.95] BY OBJECT-SIZE GROUP. POINT ESTIMATES WITH 95% PAIRED-BOOTSTRAP CONFIDENCE INTERVALS, ON A 0–100 SCALE. RELATIVE LOSSES IN THE TEXT ARE COMPUTED FROM UNROUNDED AP VALUES AND CAN DIFFER SLIGHTLY FROM RATIOS OF THE ROUNDED VALUES SHOWN HERE.

Group	Objects	FP32	INT8 seed 0
Small	50,816	6.62 [6.22, 7.19]	6.19 [5.83, 6.72]
Non-small	24,286	25.32 [24.56, 26.49]	23.93 [23.21, 25.05]
Medium	21,876	23.93 [23.19, 25.08]	22.58 [21.87, 23.68]
Large	2,410	35.84 [32.57, 42.52]	33.57 [30.59, 40.63]
All	75,102	15.44 [14.88, 16.15]	14.60 [14.06, 15.28]

All other comparisons are exploratory and are reported with confidence intervals only. These are absolute AP drops, AP@0.5, small/medium/large AP, calibration seeds 1 and 2, and the FP16 control. The INT8 result is considered directionally robust if D has the same sign for all three calibration seeds. The FP16 control shows whether a size-dependent effect is specific to INT8: a similar pattern under FP16 would prevent attributing it to INT8 alone. Exploratory intervals are not adjusted for multiplicity and are therefore not used for hypothesis testing. The implementation of training, conversion, evaluation and analysis is available in the project repository [13].

F. Experimental Setup and Deviations

All main-study training, engine builds and evaluations ran on one NVIDIA GeForce RTX 5080 (16 GB) in one fixed environment: Python 3.11.9, PyTorch 2.10.0+cu128, Ultralytics 8.4.174 and TensorRT 11.3.0.99. Ultralytics and TensorRT match the pre-registration. Python and PyTorch differ from the versions in the project's setup instructions (3.12 and 2.14.1); every run log from the first training onward shows the versions actually used. The pre-registered latency benchmark, a secondary outcome, was not implemented. As the pre-registered fallback allows, only the per-image inference times recorded by Ultralytics during evaluation are reported. Both deviations are listed in the deviation log of the pre-registration, which was completed after the experiment. Neither affects the confirmatory analysis.

The study uses only a public benchmark released for research [5]. It involves no human participants, and no images are redistributed or used to identify people.

IV. RESULTS

A. Test-Set Detection Accuracy

The final evaluation covered 1,610 test-dev images containing 75,102 retained ground-truth objects. Table I reports AP@[.5:.95] overall and for all size groups shown in Fig. 1; the non-small group is the union of the medium and large groups. Values are on a 0–100 scale, and confidence intervals are 95% paired image-bootstrap percentile intervals. INT8 seed 0 had lower point estimates than FP32 in every group. In absolute terms, INT8 lost 0.43 AP points for small objects and 1.39 points for non-small objects.

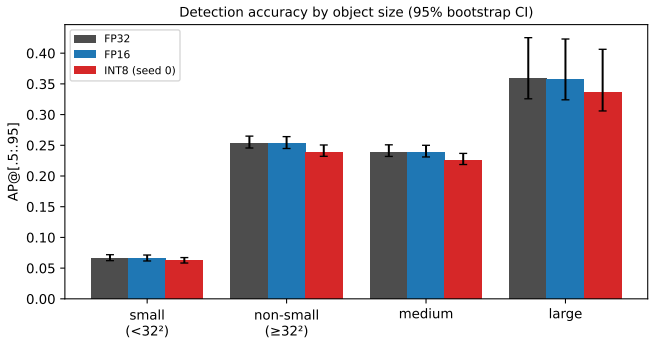


Fig. 1. Test-dev AP@[.5:.95] (0–1 scale) by object-size group for FP32, FP16 and INT8 seed 0. Error bars show 95% paired-bootstrap confidence intervals.

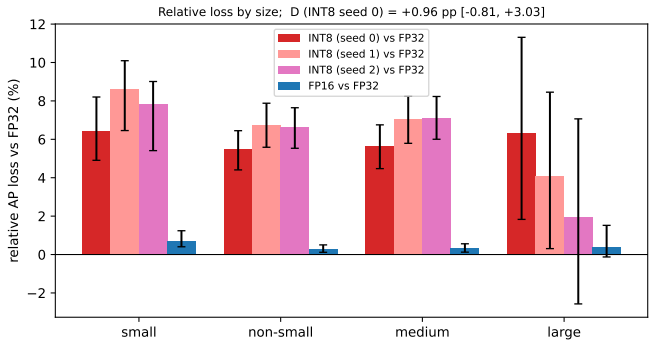


Fig. 2. Relative AP@[.5:.95] loss versus FP32 by object-size group for the three INT8 calibration seeds and FP16. Error bars show 95% paired-bootstrap confidence intervals. The primary comparison is INT8 seed 0 versus FP32.

TABLE II
RELATIVE AP LOSS VERSUS TENSORRT FP32 (%), THE SIZE CONTRAST D AND ITS ABSOLUTE COUNTERPART D_{abs} (AP POINTS), WITH 95% PAIRED-BOOTSTRAP CONFIDENCE INTERVALS, ON AP@[.5:.95] UNLESS MARKED AP50. THE FIRST ROW IS THE CONFIRMATORY TEST; ALL OTHER ROWS ARE EXPLORATORY.

Condition	Small	Non-s.	D [95% CI]	D_{abs} [95% CI]
INT8 s0	6.43	5.47	$+0.96$ $[-0.81, +3.03]$	-0.96 $[-1.24, -0.65]$
INT8 s0 (AP50)	4.57	4.87	-0.30 $[-1.74, +1.95]$	-1.42 $[-1.87, -0.79]$
INT8 s1	8.60	6.77	$+1.83$ $[-0.60, +3.72]$	-1.14 $[-1.49, -0.84]$
INT8 s2	7.84	6.65	$+1.19$ $[-1.34, +2.66]$	-1.17 $[-1.49, -0.91]$
FP16	0.70	0.28	$+0.42$ $[+0.07, +0.98]$	-0.02 $[-0.08, +0.03]$

B. Confirmatory Size-Dependent Loss

For INT8 seed 0, the relative AP loss was 6.43% for small objects and 5.47% for non-small objects (Fig. 2). Their difference was $D = +0.96$ percentage points, with a 95% confidence interval of $[-0.81, +3.03]$ percentage points. Because the interval contains zero, the pre-registered result is inconclusive and does not support the hypothesis that INT8 causes a larger relative AP loss for small objects. Fig. 3 shows the bootstrap distribution of the primary contrast.

C. Exploratory Comparisons

Table II summarizes the exploratory contrasts. The point estimate of D was positive for all three INT8 calibration seeds ($+0.96$, $+1.83$ and $+1.19$ percentage points), which

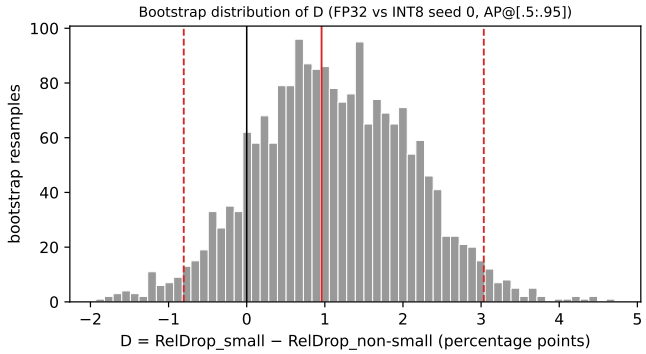


Fig. 3. Paired-bootstrap distribution of D for the primary FP32 versus INT8 seed 0 comparison. The solid line marks the point estimate, the dashed lines the 95% confidence limits, and the zero line no difference in relative loss by size.

meets the pre-registered criterion for directional robustness. All three intervals included zero, however, so the confirmatory finding remains inconclusive. The overall relative loss varied more between seeds (5.4%, 7.4% and 7.0%) than D did. On AP@0.5, D was close to zero for every seed. On the absolute scale, D_{abs} was negative for every seed with intervals excluding zero: in AP points, small objects lost less than non-small objects.

The FP16 engine lost only 0.37% of overall AP, which shows that TensorRT conversion itself costs little accuracy and that the INT8 loss comes from quantization. Its D was $+0.42$ percentage points with an interval just above zero, although D_{abs} was essentially zero. Because the primary INT8 interval includes zero, the pre-registered FP16 attribution rule is not triggered.

D. Assumption and Implementation Checks

All 2,000 resamples produced finite AP values for every condition and size group. The bootstrap distribution of D (Fig. 3) is unimodal and nearly symmetric (skewness 0.06), with a standard error of 0.99 percentage points. Its mean exceeds the point estimate by only 0.16 points. A normal-approximation interval (point estimate ± 1.96 standard errors, $[-0.99, +2.91]$) also contains zero, so the decision does not depend on the interval method. The implementation's AP matched pycocotools to within 2×10^{-16} for every engine. TensorRT FP32 and PyTorch FP32 differed by less than 10^{-4} AP@[.5:.95] in every size group, so the TensorRT FP32 engine is a faithful baseline. Overall AP differed from Ultralytics' own validator by 0.0028–0.0029 for every TensorRT engine. The offset reflects a different AP implementation; it is nearly constant across conditions and therefore does not affect the paired contrasts.

E. Latency

The controlled latency benchmark was not run (Section III). The per-image inference times that Ultralytics recorded during test-dev evaluation were 0.77 ms for FP32, 0.42 ms for FP16

and 0.69–0.71 ms for the three INT8 engines. These times come from an uncontrolled protocol and are descriptive only.

V. Discussion

The results do not provide sufficient evidence that INT8 post-training quantization causes a greater relative detection-accuracy loss for small objects than for non-small objects. The relative loss was higher for small objects (6.43%) than for non-small objects (5.47%), but the difference of 0.96 percentage points had a 95% confidence interval spanning zero. The primary hypothesis was therefore not supported. This finding is inconclusive; it is not evidence that the two groups are affected equally. The interval does, however, bound the effect. A size-dependent penalty larger than about 3 percentage points, meaning a small-object relative loss above roughly 8.5% against 5.5% for non-small objects, is not compatible with the data. Compared with the overall INT8 loss of 5.4%, any size-dependent component is therefore at most moderate.

The relative and absolute views point in different directions. Small objects lost proportionally slightly more, whereas non-small objects lost about three times more AP points (1.39 vs. 0.43). Statements that quantization "hurts small objects more" thus depend on the scale used. The relative scale was pre-registered because the baselines differ by a factor of about four. The mechanisms proposed in the introduction also act on different scales. Coarser activations would tend to make small objects disappear, which should show up at IoU 0.5. Localization jitter would mainly hurt small objects at high IoU thresholds. That D was near zero on AP@0.5 but slightly positive on AP@[.5:.95] is weakly consistent with the second mechanism rather than the first, but the difference is well within the uncertainty.

The positive estimates across all three calibration seeds indicate a consistent direction, but the seeds share the same model and test images and are not independent replications. The choice of calibration set changed the overall INT8 loss by two percentage points, which is larger than the size contrast itself. The FP16 control lost almost no accuracy overall, yet produced a small positive D with an interval just above zero. Because the small-object FP16 loss amounts to only 0.05 AP points, this relative difference is fragile. It may indicate that very small objects are sensitive to any reduction in precision, or it may be a chance finding among several uncorrected exploratory intervals. Outlier-sensitive calibration, as studied by Kim et al. [8], is one possible mechanism for differences between calibration sets; this experiment does not test that mechanism.

A. Threats to Validity

Internal validity. All conditions share one checkpoint, one ONNX graph, identical preprocessing and identical images, so precision is the only manipulated factor. TensorRT does, however, choose kernels and layer fusions per precision, and the INT8 engine runs its unquantized layers in FP16. The FP16 control bounds this toolchain effect but cannot fully separate it from quantization.

Construct validity. Object size is observed, not assigned. Small and non-small objects differ in class composition (Section III), so D measures size and class together. Small objects also start from a low AP of 0.066, which makes relative losses noisy. The excluded ignored regions mean that absolute AP differs from the official VisDrone protocol [6], although the same rule applies to every condition. Latency was not measured under a controlled protocol. The recorded times show no speed-up of INT8 over FP16 at batch size 1, so the study cannot show that the accuracy loss buys faster inference on this GPU.

Conclusion validity. The bootstrap captures variation across test images but not variation from model training or hardware. Correlation between images of the same scene would make its intervals too narrow, which would make a significant result less trustworthy. It does not explain the inconclusive one. The CI width of about 4 percentage points means the study could detect only a moderate size effect. Finally, the pre-registration was not deposited in a time-stamped registry, and the public repository was first committed after the test-dev analysis. Its date therefore rests on the authors' own record.

External validity. The experiment uses one YOLOv8n model, one dataset and a 640×640 full-image input, which downscales the original images. Slicing-aided inference [7] enlarges small objects in the input, so the findings should not be generalized to tiled pipelines, larger detectors or other quantization toolchains.

VI. Conclusion

This study investigated whether INT8 post-training quantization causes a greater relative detection-accuracy loss for small objects than for non-small objects in aerial imagery. A fine-tuned YOLOv8n detector was evaluated as TensorRT FP32, FP16 and INT8 engines on the VisDrone2019-DET test-dev split, using size-specific AP and a paired image-level bootstrap.

For the primary FP32-versus-INT8 comparison, the relative AP loss was 6.43% for small objects and 5.47% for non-small objects. The difference of 0.96 percentage points had a 95% confidence interval of $[-0.81, +3.03]$ percentage points. Since the interval includes zero, the analysis did not establish that small objects suffer a greater relative loss. The consistent direction across three calibration seeds did not resolve this uncertainty. In absolute AP points, small objects lost significantly less than non-small objects. The FP16 control lost almost no accuracy, so the overall loss of 5–7% is caused by INT8 quantization, but the size pattern is not necessarily specific to INT8.

Overall, the experiment provides no conclusive evidence for a size-dependent INT8 accuracy loss under the evaluated conditions. Future work should test additional models and datasets and measure inference performance under a controlled protocol. Size and class could be separated by estimating D within a single class, such as cars, which are numerous in both groups (15,112 small and 12,962 non-small), or by rescaling test images so that the same objects cross the size threshold.

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